

A Zero-Preserving, Singularity-Sensitive Division Calculus with an Idempotent Element

Manu R

September 14, 2026

Abstract

This paper proposes a division calculus in which division by zero is not interpreted as undefined, but neither is it treated as ordinary division by an invertible element. A distinguished algebraic element I is introduced through the division-calculus resolution of $1/0$, while multiplication remains zero-preserving: $0x = x0 = 0$. The element I is idempotent, $I^2 = I$, and division by I resolves to multiplication by zero.

The central feature of the proposal is that division is treated as a syntax-sensitive computational operation rather than as an unrestricted binary operation on algebraic values. In particular, denominators containing 0 or I are resolved by explicit singular-factor rules before ordinary algebraic simplification is permitted. Chained division is right-associated and evaluated from the innermost division outward. Singular factors generated by an inner division must be retained when that division expression occurs inside an outer denominator.

Consequently, familiar algebraic transformations such as cancellation, arbitrary denominator simplification, and unrestricted reassociation of division are not valid in the singular cases. The resulting framework preserves ordinary arithmetic away from 0 and I while providing a consistent operational treatment of expressions involving these two distinguished elements.

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1 Introduction

Division by zero is traditionally excluded from ordinary arithmetic. This exclusion is natural within fields because division is defined through multiplicative inverses, and zero has no multiplicative inverse.

This paper explores a different approach.

Rather than declaring every expression involving division by zero to be undefined, we introduce a distinguished element I associated with the division expression

$$\frac{1}{0}.$$

The crucial point is that this association does *not* turn 0 into an ordinary invertible element. Instead, the notation $1/0$ is a special division expression that resolves to a separate algebraic value I .

The framework therefore separates two concepts:

1. an underlying algebra of values containing 0, 1, and I ;
2. a division calculus that specifies how division expressions are resolved.

This distinction is essential. Ordinary algebraic identities continue to apply to resolved values, but they cannot be used indiscriminately to rewrite an unresolved singular denominator.

For example, the framework intentionally permits

$$\frac{1}{0} = I$$

while retaining

$$I0 = 0.$$

Thus I is not the ordinary multiplicative inverse of zero.

Similarly,

$$\frac{I}{I} = 0$$

rather than 1.

The purpose of this paper is to define these rules precisely, demonstrate their consequences, and identify the algebraic transformations that must be restricted in the presence of singular denominators.

2 Underlying Algebra

2.1 The distinguished element

We introduce a distinguished element I satisfying

$$I^2 = I.$$

Thus I is idempotent.

More generally,

$$I^n = I, \quad n \geq 1.$$

The ordinary multiplication rule involving zero remains unchanged:

$$0x = x0 = 0$$

for every algebraic value x , including

$$0I = I0 = 0.$$

These multiplication rules are fundamental. In particular, the introduction of I does not alter the zero-preserving property of multiplication.

2.2 The underlying algebra

One possible algebraic carrier for the values is

$$A = \mathbb{R}[I]/(I^2 - I).$$

Every element can be represented as

$$a + bI, \quad a, b \in \mathbb{R}.$$

Addition is defined componentwise:

$$(a + bI) + (c + dI) = (a + c) + (b + d)I.$$

Multiplication follows from $I^2 = I$:

$$(a + bI)(c + dI) = ac + (ad + bc + bd)I.$$

There is a useful representation

$$a + bI \longmapsto (a, a + b)$$

which identifies this algebra with

$$\mathbb{R} \times \mathbb{R}.$$

Under this representation,

$$I \longmapsto (0, 1).$$

Consequently,

$$I(1 - I) = 0.$$

Thus I is a zero divisor. This is not a defect of the construction; it is an intentional property of the underlying algebra.

3 Division Is a Separate Calculus

The central principle of this proposal is:

Division is a computational operation with a restricted ordinary domain, not an unrestricted binary operation on all algebraic values.

For an ordinary denominator b satisfying

$$b \notin \{0, I\},$$

division retains its ordinary meaning.

When the denominator is 0 or I , ordinary division is not invoked. Instead, the division expression is resolved using explicit singular rules.

This distinction is important because the notation

$$\frac{1}{0} = I$$

must not be interpreted as the statement that I is an ordinary multiplicative inverse of 0.

It is a division-calculus rule.

3.1 Resolution notation

To emphasize the distinction between ordinary equality and division resolution, we may write

$$E \rightsquigarrow V$$

to mean that the division expression E resolves to the algebraic value V .

Thus

$$\frac{1}{0} \rightsquigarrow I$$

means that the division expression $1/0$ resolves to the value I .

It does not mean that ordinary multiplicative inversion has been extended to zero.

4 Fundamental Singular Division Rules

The fundamental rules are:

Rule 4.1 (Division by zero). *For every algebraic numerator a ,*

$$\frac{a}{0} \rightsquigarrow aI.$$

In particular,

$$\frac{1}{0} \rightsquigarrow I$$

and

$$\frac{0}{0} \rightsquigarrow 0I = 0.$$

Rule 4.2 (Division by I). *For every algebraic numerator a ,*

$$\frac{a}{I} \rightsquigarrow a0.$$

Therefore,

$$\frac{1}{I} \rightsquigarrow 0,$$

$$\frac{I}{I} \rightsquigarrow I0 = 0,$$

and

$$\frac{0}{I} \rightsquigarrow 0.$$

These rules can be viewed as arising from the reciprocal-resolution identities

$$\frac{1}{0} \rightsquigarrow I, \quad \frac{1}{I} \rightsquigarrow 0.$$

They are resolution rules, not ordinary inverse identities.

5 Zero-Preserving Multiplication

The introduction of I does not modify multiplication by zero.

For all values x ,

$$0x = x0 = 0.$$

Hence

$$0I = I0 = 0.$$

This has an important consequence:

$$\left(\frac{1}{0}\right) 0 \rightsquigarrow I0 = 0.$$

It is therefore *not* valid to apply the familiar inverse-cancellation intuition

$$\left(\frac{1}{0}\right) 0 = 1.$$

The expression $1/0$ has resolved to the value I ; it has not established that I is a multiplicative inverse of zero.

6 Singular Factors in Denominators

A central rule of the calculus is that singular factors in a denominator must be resolved before ordinary multiplication or simplification of those factors is performed.

6.1 A zero factor

For an ordinary nonzero c ,

$$\frac{x}{0c} \rightsquigarrow \frac{xI}{c}.$$

The zero factor contributes I to the numerator.

Importantly, one must not first apply the ordinary algebraic identity

$$0c = 0$$

and then replace the denominator by 0.

Doing so would lose the factor structure that the division calculus uses.

Thus, in general,

$$\frac{x}{0c} \neq \frac{x}{0}.$$

The right-hand side is

$$xI,$$

whereas the factor-sensitive expression gives

$$\frac{xI}{c}.$$

6.2 An I factor

Similarly, for ordinary nonzero c ,

$$\frac{x}{Ic} \rightsquigarrow \frac{x0}{c} = 0.$$

Thus a denominator factor I contributes a factor 0 to the numerator.

6.3 Multiple singular factors

For multiple zero factors,

$$\frac{x}{00c} \rightsquigarrow \frac{xII}{c} = \frac{xI}{c}.$$

The idempotence relation

$$I^2 = I$$

then applies after the singular factors have been transferred.

Likewise,

$$\frac{x}{IIc} \rightsquigarrow \frac{x00}{c} = 0.$$

7 Right-Associated Division

Chained division is interpreted from the bottom upward, or equivalently, right-associatively.

Thus

$$a/b/c$$

means

$$\frac{a}{b/c},$$

not

$$\frac{a/b}{c}.$$

This convention is particularly important in the presence of singular values.

7.1 Example: $1/0/0$

Consider

$$1/0/0.$$

By the right-association rule,

$$1/0/0 \equiv \frac{1}{0/0}.$$

The inner division generates the singular-factor representation

$$0/0 \rightsquigarrow 0I.$$

Because this expression occurs as the denominator of an outer division, its singular-factor structure must be retained until the outer denominator is resolved:

$$\frac{1}{0/0} = \frac{1}{0I}.$$

The two singular denominator factors are transferred to the numerator:

$$\frac{1}{0I} \rightsquigarrow I0 = 0.$$

Therefore,

$$\boxed{1/0/0 = 0}.$$

This must not be transformed into

$$(1/0)/0.$$

The latter is a different expression and is not the interpretation of the unchained notation $1/0/0$.

7.2 Example: $I/0$

Separately,

$$\frac{I}{0} \rightsquigarrow II = I.$$

Thus

$$\boxed{I/0 = I}$$

while

$$\boxed{1/0/0 = 0}.$$

There is no contradiction because these are different division expressions.

8 Division by I

The same principles apply to I .

For example,

$$\frac{I}{I} \rightsquigarrow I0 = 0.$$

Thus

$$\boxed{I/I = 0}.$$

There is no cancellation of the two occurrences of I .

Likewise,

$$\frac{1}{I} \rightsquigarrow 0.$$

The expression I/I is therefore fundamentally different from ordinary expressions a/a with an ordinary nonzero a .

9 Cancellation Rules

Ordinary cancellation is restricted to the ordinary domain.

For an ordinary nonzero a ,

$$\frac{a}{a} = 1.$$

However,

$$\frac{0}{0} = 0$$

and

$$\frac{I}{I} = 0.$$

Consequently, the rule

$$\frac{a}{a} = 1$$

cannot be extended to $a = 0$ or $a = I$.

Similarly,

$$\left(\frac{a}{0}\right) 0 = aI0 = 0,$$

not a .

The same applies to cancellation through I :

$$\left(\frac{a}{I}\right) I = a0I = 0.$$

Therefore, neither 0 nor I may be cancelled in the manner of an ordinary nonzero denominator.

10 Why Ordinary Denominator Simplification Is Restricted

The framework deliberately distinguishes algebraic equality from rewriting inside an unresolved division expression.

For example, the underlying algebra satisfies

$$0 = 2 \cdot 0.$$

However, it does not follow that

$$\frac{1}{0} = \frac{1}{2 \cdot 0}.$$

The left side resolves to

$$I,$$

while the factor-sensitive right side resolves to

$$\frac{I}{2}.$$

Therefore,

$$\frac{1}{0} \neq \frac{1}{2 \cdot 0}$$

under the division calculus.

This is not a contradiction in the underlying algebra. It means that ordinary equality of algebraic values is not an unrestricted rewrite rule inside an unresolved singular denominator.

The denominator carries structural information relevant to division resolution.

11 Nested Division

Nested division must preserve singular structure until the containing division has been resolved.

Consider

$$\frac{x}{y/0},$$

where y is an ordinary nonzero value.

The inner division resolves to

$$\frac{y}{0} \rightsquigarrow yI.$$

Therefore the outer denominator becomes

$$yI.$$

The singular factor I is transferred to the numerator as 0:

$$\frac{x}{yI} \rightsquigarrow \frac{x0}{y} = 0.$$

Hence

$$\boxed{\frac{x}{y/0} = 0} \quad (y \neq 0).$$

The same structural principle applies when the inner division itself contains multiple singular factors.

The important rule is that an inner division expression cannot first be collapsed to an ordinary algebraic value if doing so would erase singular factor information needed by an outer denominator.

12 Context-Sensitive Evaluation

The preceding rules imply that the calculus is not purely compositional with respect to the ordinary algebraic value of every subexpression.

For example, as a standalone expression,

$$\frac{0}{0} \rightsquigarrow 0I = 0.$$

But when the expression occurs inside an outer denominator,

$$\frac{1}{0/0},$$

the singular-factor representation

$$0I$$

must be preserved:

$$\frac{1}{0/0} = \frac{1}{0I} \rightsquigarrow I0 = 0.$$

If the inner expression were first replaced by its normalized value 0, one would obtain

$$\frac{1}{0} = I,$$

which is not the prescribed result.

Therefore the calculus distinguishes between:

1. the normalized algebraic value of a completed expression; and
2. the structured representation of a division expression while it remains part of a denominator.

This distinction is a fundamental part of the proposed calculus.

13 Forbidden Transformations

The following transformations are not generally valid when singular denominators are involved.

13.1 Cancellation of singular factors

Invalid:

$$\frac{I}{I} = 1.$$

Correct:

$$\frac{I}{I} = 0.$$

13.2 Cancellation through zero

Invalid:

$$\left(\frac{a}{0}\right) 0 = a.$$

Correct:

$$\left(\frac{a}{0}\right) 0 = aI0 = 0.$$

13.3 Replacing a singular denominator by an algebraically equal expression

Invalid:

$$\frac{1}{0} = \frac{1}{2 \cdot 0}.$$

The two denominators have different factor structures under the division calculus.

13.4 Reassociating chained division

Invalid:

$$1/0/0 = (1/0)/0.$$

The prescribed interpretation is

$$1/0/0 = 1/(0/0).$$

13.5 Premature normalization

Invalid:

$$\frac{1}{0/0} \rightarrow \frac{1}{0} \rightarrow I.$$

The prescribed evaluation is

$$\frac{1}{0/0} \rightarrow \frac{1}{0I} \rightarrow I0 \rightarrow 0.$$

13.6 Ordinary fraction cancellation

The familiar identity

$$\frac{a/b}{c/d} = \frac{ad}{bc}$$

cannot be applied without checking that all relevant denominators and intermediate division expressions lie within the ordinary division domain.

The singular calculus takes precedence over such transformations.

14 Ordinary Arithmetic Is Preserved

The proposal does not seek to replace ordinary arithmetic.

If all denominators encountered during evaluation are ordinary nonzero values, the standard rules of arithmetic apply.

For example, if

$$a, b, c \neq 0$$

and all intermediate expressions remain in the ordinary domain, then standard identities such as

$$\frac{a}{b} \frac{b}{c} = \frac{a}{c}$$

remain valid.

The special rules are activated only when 0 or I enters a denominator.

This gives the calculus a natural boundary:

Ordinary denominator	ordinary division rules
Denominator 0	transfer $0 \mapsto I$ to numerator
Denominator I	transfer $I \mapsto 0$ to numerator
Nested division	preserve singular structure until outer resolution

15 Basic Resolution Table

For the singular numerators and denominators, the rules give:

$$\begin{array}{c|cc} / & 0 & I \\ \hline 0 & 0 & 0 \\ I & I & 0 \end{array}$$

Specifically,

$$\begin{array}{ll} \frac{0}{0} = 0, & \frac{0}{I} = 0, \\ \frac{I}{0} = I, & \frac{I}{I} = 0. \end{array}$$

This table should not be interpreted as ordinary field division. It is a summary of the singular division-resolution rules.

16 Relationship Between the Two Layers

The framework can therefore be viewed as having two distinct mathematical layers.

16.1 Value layer

The value layer contains algebraic objects such as

$$0, 1, I, a + bI$$

with ordinary addition and multiplication satisfying

$$I^2 = I, \quad 0x = 0.$$

16.2 Division-expression layer

The expression layer determines how notation involving ‘/’ is resolved.

The division layer has its own structural rules concerning:

- the ordinary domain of division;
- singular denominators;
- factor-sensitive denominator processing;
- right-associated chained division;
- preservation of singular structure in nested denominators.

The two layers should not be conflated.

In particular, the statement

$$1/0 \rightsquigarrow I$$

belongs to the division layer, while

$$I0 = 0$$

belongs to the value layer.

There is no rule that allows the former to be interpreted as an ordinary inverse identity.

17 Conceptual Interpretation

The proposal can be summarized by the following principle:

A fraction is a representation of a division operation. Division is ordinary only when its denominator is outside the singular set $\{0, I\}$. When a singular denominator occurs, the expression is resolved using explicit singular-factor rules rather than ordinary inverse arithmetic.

Under this interpretation, I is not “infinity,” nor is it an ordinary multiplicative inverse of zero.

It is a distinguished algebraic value associated with the singular division expression $1/0$.

The zero-preserving rule

$$0I = 0$$

remains intact.

This makes the framework fundamentally different from approaches in which division by zero is interpreted as an absorbing infinity-like value or in which zero is assigned an ordinary multiplicative inverse.

18 Examples

18.1 Basic examples

$$\frac{1}{0} = I$$

$$\frac{1}{I} = 0$$

$$\frac{0}{0} = 0$$

$$\frac{0}{I} = 0$$

$$\frac{I}{0} = I$$

$$\frac{I}{I} = 0.$$

18.2 Products

$$\left(\frac{1}{0}\right) 0 = I0 = 0.$$

$$\left(\frac{1}{0}\right) I = I^2 = I.$$

18.3 Factor-sensitive denominators

For $c \neq 0$,

$$\frac{x}{0c} = \frac{xI}{c}.$$

For $c \neq 0$,

$$\frac{x}{Ic} = \frac{x0}{c} = 0.$$

18.4 Chained division

$$1/0/0 = \frac{1}{0/0} = \frac{1}{0I} = I0 = 0.$$

18.5 Nested singular division

For ordinary nonzero y ,

$$\frac{x}{y/0} = \frac{x}{yI} = \frac{x0}{y} = 0.$$

19 Consistency Requirements

The proposal should not be understood as claiming that every familiar identity of field arithmetic remains valid.

Instead, consistency requires that the permitted rewrite system respect the division-expression rules.

In particular, the following principles are required:

1. A singular denominator must be processed before ordinary denominator simplification.
2. A singular factor generated by an inner division must be retained when that division expression occurs inside an outer denominator.
3. Division expressions are right-associated unless explicitly parenthesized otherwise.
4. Parentheses are semantically significant in division expressions.
5. Cancellation is permitted only for ordinary nonzero denominators.
6. Equality of resolved algebraic values does not automatically license substitution inside an unresolved division expression.
7. The relations

$$1/0 \rightsquigarrow I, \quad 1/I \rightsquigarrow 0$$

are division-resolution rules and must not be interpreted as ordinary multiplicative inverse equations.

These restrictions prevent the standard algebraic transformations from producing conflicting results.

20 Discussion

The principal novelty of the proposed framework is not merely the introduction of an idempotent element. Idempotent extensions of algebraic systems are well known, as are several mathematical frameworks that assign special meanings to division-by-zero expressions.

The distinctive feature proposed here is the combination of:

1. a distinguished idempotent element I ;
2. preservation of ordinary zero multiplication,

$$0I = 0;$$

3. explicit singular division resolution,

$$a/0 \rightsquigarrow aI, \quad a/I \rightsquigarrow a0;$$

4. factor-sensitive treatment of singular denominators;
5. right-associated, bottom-up evaluation of chained division;
6. preservation of singular-factor structure through nested denominators; and
7. explicit rejection of ordinary cancellation and denominator rewriting in singular contexts.

The resulting object is therefore better viewed as a *division calculus over an algebraic carrier* than as a conventional field-like structure equipped with a totalized binary division operation.

21 Open Formal Questions

Several questions remain for a complete formal development.

21.1 Formal grammar

A rigorous grammar should distinguish ordinary algebraic expressions from division expressions whose denominator remains unresolved.

21.2 Operational semantics

The resolution rules should be given as an ordered or context-sensitive rewriting system so that expressions such as

$$\frac{1}{0/0}$$

have exactly the intended evaluation path.

21.3 Confluence

Because ordinary algebraic simplification is restricted during division resolution, the resulting rewrite system should be studied for confluence under its permitted rules.

21.4 Normal forms

A formal normal-form representation should specify when singular factors are retained and when ordinary algebraic normalization is allowed.

21.5 Equational theory

The system should distinguish equations between resolved values from rewriting rules applicable to unresolved division expressions.

These distinctions are essential for determining precisely which familiar algebraic identities survive and which do not.

22 Conclusion

This paper proposes a zero-preserving division calculus built around a distinguished idempotent element I .

The essential definitions are

$$I^2 = I,$$

$$0x = x0 = 0,$$

$$\frac{a}{0} \rightsquigarrow aI,$$

and

$$\frac{a}{I} \rightsquigarrow a0.$$

The central conceptual distinction is that I is an algebraic value produced by a special division-resolution rule; it is not the ordinary multiplicative inverse of zero.

Division is therefore not treated as an unrestricted binary operation. Expressions containing singular denominators are evaluated according to a specific syntax-sensitive calculus. Singular factors are transferred from denominator to numerator, ordinary simplification is delayed until the singular structure has been resolved, and chained division is evaluated bottom-up/right-associatively.

Under these rules,

$$\frac{1}{0} = I, \quad \frac{1}{I} = 0, \quad \frac{0}{0} = 0, \quad \frac{I}{I} = 0,$$

while

$$1/0/0 = \frac{1}{0/0} = \frac{1}{0I} = I0 = 0.$$

The framework therefore does not attempt to preserve all identities of ordinary field arithmetic. Instead, it defines a different computational semantics for division while retaining the familiar zero-preserving multiplication and a well-defined algebraic carrier.

A complete mathematical treatment requires a formal syntax, operational semantics, normal-form theory, and a proof of consistency/confluence for the resulting rewrite system. Those developments constitute the natural next stage of the theory.